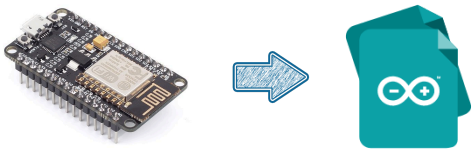
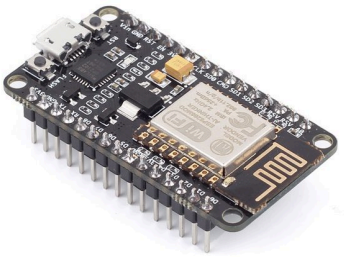


NODEMCU-a PROGRAMATZEN

BEHARREZKOAK

- ➢ **NodemCU V1.0**
- ➢ **Arduino IDE 2.3.6** https://www.arduino.cc/en/software/



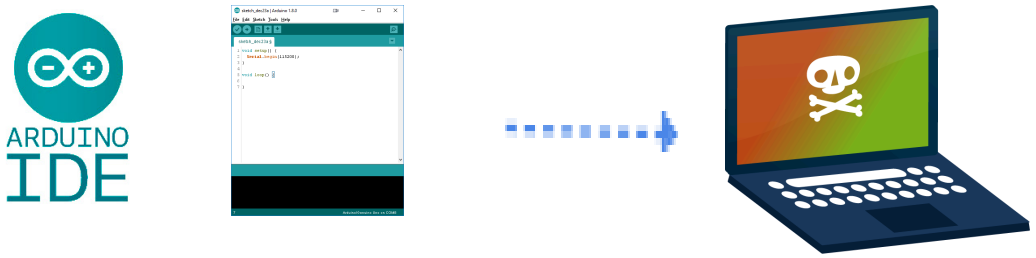
1) Arduino IDEaren instalazioa.

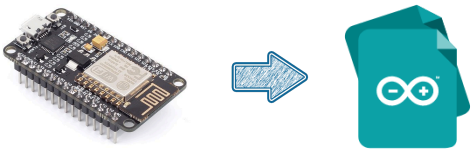
NodeMCUa ESP8266an oinarritutako programazio plaka izaki eLUAn programatu daiteke, baita MicroPythonen ere baino guk Arduino lenguaian programatuko dugu gure kurtsoan zehar.

Horretarako Arduino IDEa instalatu beharko dugu gure ordenagailuan. Hau daukagun sistema eragilearen arabera desberdin egin daiteke beraz jarraitu arduino.cc orrialdean ematen dizkizuen instalazio gomendioak.

Gogoratu uneko azken bertsio egonkorra instalatzea. Momentu honetan azkenengoa **2.3.6** a da.

www.arduino.cc (**IDE Downloads**)
www.arduino.cc (**IDE Instalation guide**)



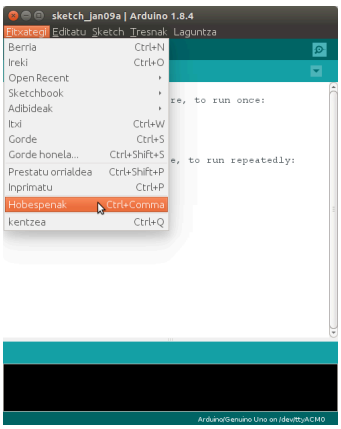


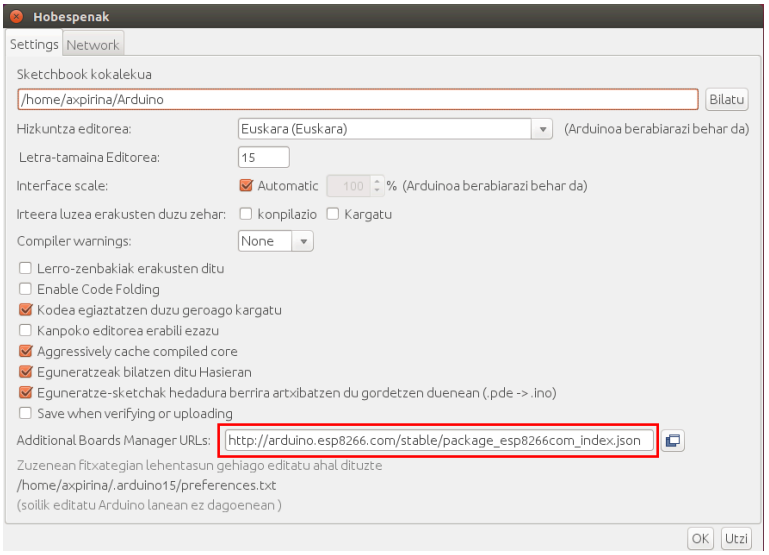
a) Arduino IDEan NodeMCU plaka instalatu.

NodeMCU plaka instalatzen ez den bitartean Arduino IDEa ez da gai izango NodeMCU plakarik ezagutzeko eta ezin izango dugu beraz programarik bidali. Hurrengo pausuak jarraitu NodeMCUa Arduino IDEan instalatzeko garaian.

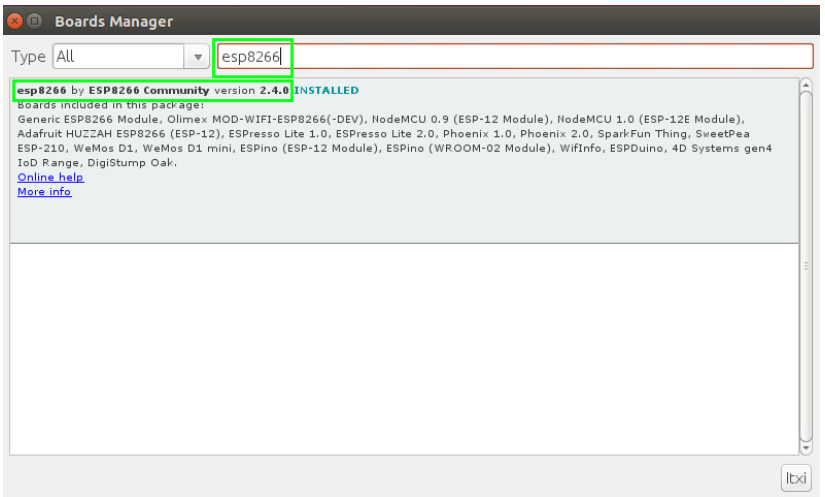
- **Fitxategiak** -> **Hobespenak**-era joan eta hurrengo esteka itsatsi **Additional Boards manager URL** kutxatilan.

http://arduino.esp8266.com/stable/package_esp8266com_index.json

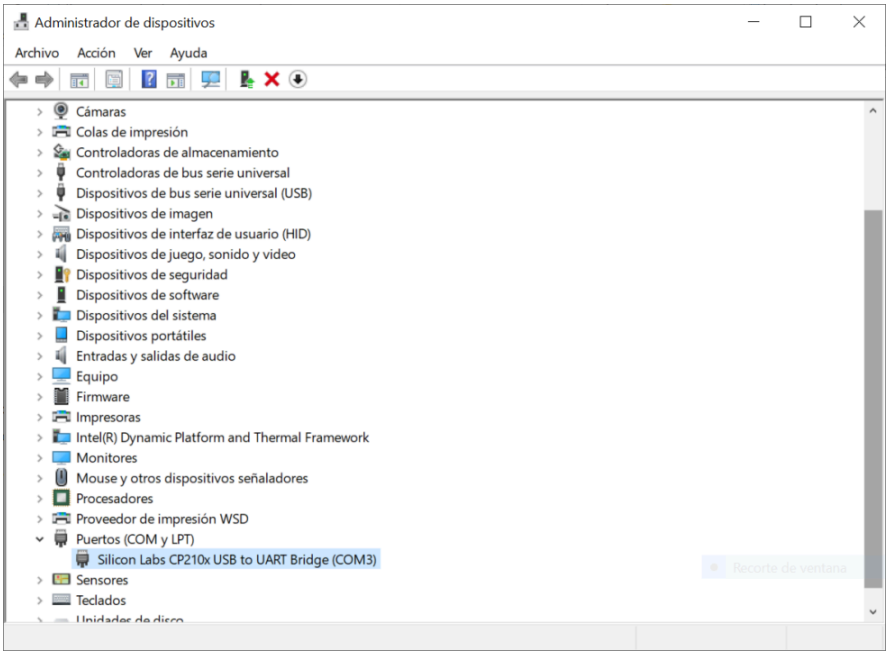




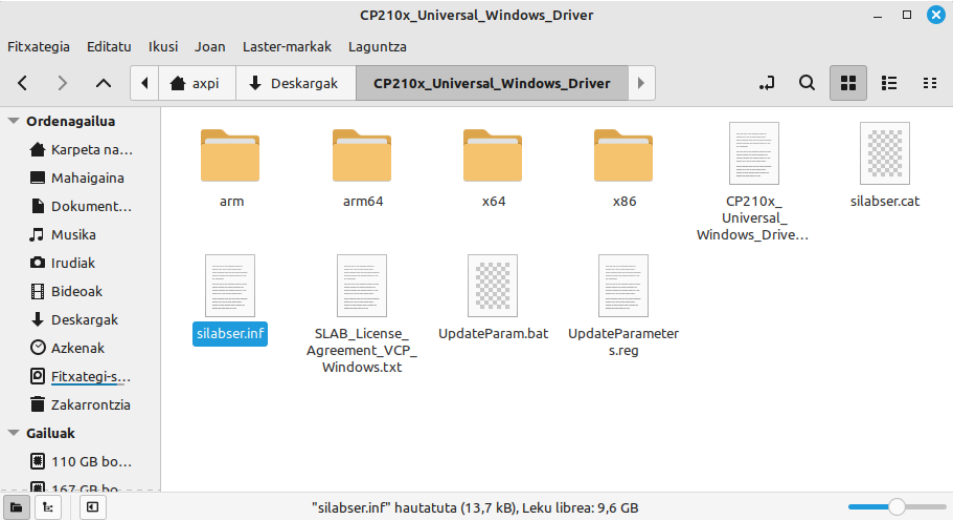
- **Tresnak** -> **Plaka -> Boards Manager**-era joan eta **ESP8266** mikroa duten plakak bilatu eta instalatu.



- **OHARRA**: Aurreko pausuekin nahikoa ez bada eta Windows 11 bat baldin badaukazu ziurrenik CP210X driverrak gaizki instalatuta dauzkazu.
    a) Desinstalatu aurreko Driverrak:
    Windows-en **Gailu Kudeatzailera** joan (**Win + X eta Gailu Kudeatzaila**) eta akats ikurra duen **COM** gailuaren gainean xaguaren eskuineko botoia sakatuz kontrolatzailea ezabatu.



    b) Joan **Silabs** etxeko orrialdera eta instalatu [CP210x Universal Windows Driver](#) jaitsi, deskonprimatu eta **silabser.inf** artxiboaren gainen eskuineko botoiarekin klik egin eta **Instalatu**.

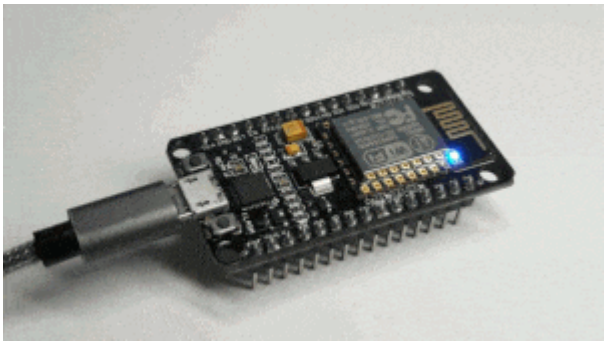


PROGRAMAK (Errazetik zailera)

NodeMCUak txertaturik dituen LEDa eta BOTOIA erabiliko ditugu gure lehen scriptak programatu nahiz probatzeko. Gogoratuko dugu **LED**a **GPIO02**an ekartzen duela konektatua eta **BOTOIA** berriz **GPIO00**an ekartzen duela konektatua.

BLINK (Dir-dir aplikazioa)

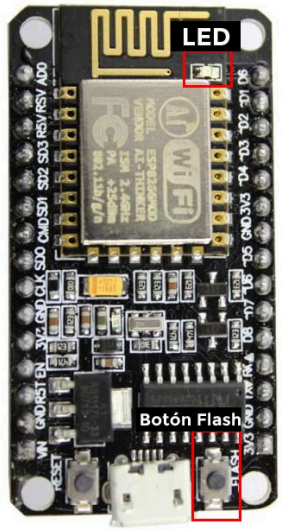
Hurrengo kodea itsatsi eta NodeMCUari bidaliaz gero NodeMCUak txertaturik duen LEDa dir-dirka jarriko. Hau da izango da gure plaka ondo dabilenaren eta programa ongi irauli dugunaren erakusle nagusia.

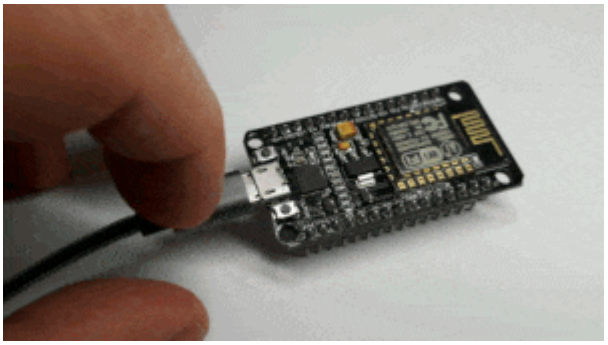


```
// setup funtzioa behin bakarrik exekutatzen da, plaka piztean edo berrabiaraztean

void setup() {
  // LED_BUILTIN pina irteera pin deklaratu
  pinMode(LED_BUILTIN, OUTPUT);
}

// loop funtzioa behin eta berriz errepikatzen da. Programa nagusia hemen itsasten
da

void loop() {
  digitalWrite(LED_BUILTIN, HIGH); // LEDa piztu voltaia 5 bihurtuz HIGH
  delay(1000); // segundu 1 itxaron
  digitalWrite(LED_BUILTIN, LOW); // LEDa itzaldu voltaia 0 bihurtuz LOW
  delay(1000); // segundu 1 itxaron
}
```

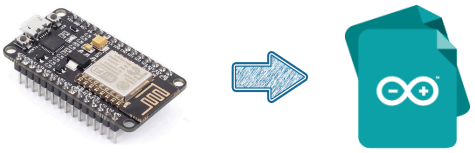
BOTOIA (Flash botoia)

NodeMCUak firmwarea kargatzeko Flash botoi bat ekartzen du. Botoi honetaz baliatuko gara gure lehen sarrera digitala erabiliz eginiko programa martxan jartzeko. Kontuan izan programa kargatu harte ezingo dugula botoirik zanpatu programa iraulita eta exekuzioan izan harte.





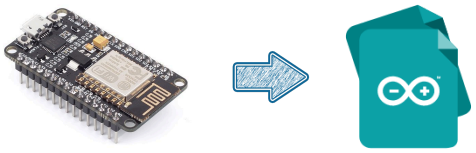
```
#define LED_BUILTIN 2
#define BUTTON_BUILTIN 0

void setup() {
  // LEDa 2. pinera itsatsia dago
  pinMode(LED_BUILTIN, OUTPUT);
  pinMode(BUTTON_BUILTIN, INPUT);
}

void loop() {
  // Botoiaren egoera irakurri. Sakatua dagoenean baxu jartzen da
  int egoera = digitalRead(BUTTON_BUILTIN);

  // LEDa konfiguratu botoiaren arabera pizteko
  digitalWrite(LED_BUILTIN, egoera);
}
```



OTEITZA-ko Wifi Sarera konektatu

NodeMCUa Oteitzako Wifi sarera konektatzen ikasiko dugu programa honekin.

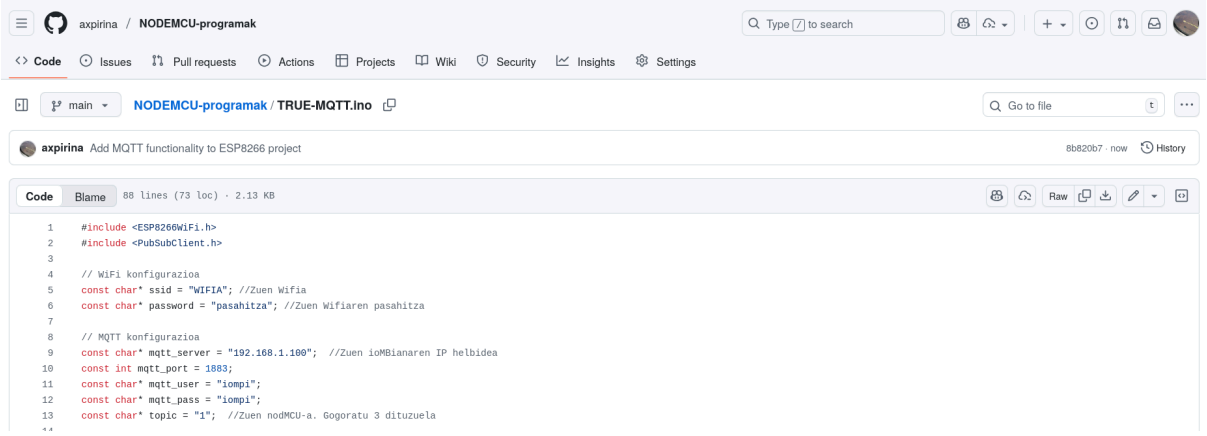
```
#include <ESP8266WiFi.h> // WiFi library for ESP8266

// WiFi  configuration
const char* ssid = "OLP";    // OLP
const char* password = "oteitzaLP";   // oteitzaLP

// Funtzioak
void setup_wifi() {
  delay(10);
  Serial.println();
  Serial.print("Wifi-ra konektatzen: ");
  Serial.println(ssid);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("");
  Serial.println("WiFira konektatua !!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
}

void setup() {
  Serial.begin(9600);
  setup_wifi();
}

void loop() {
}
```

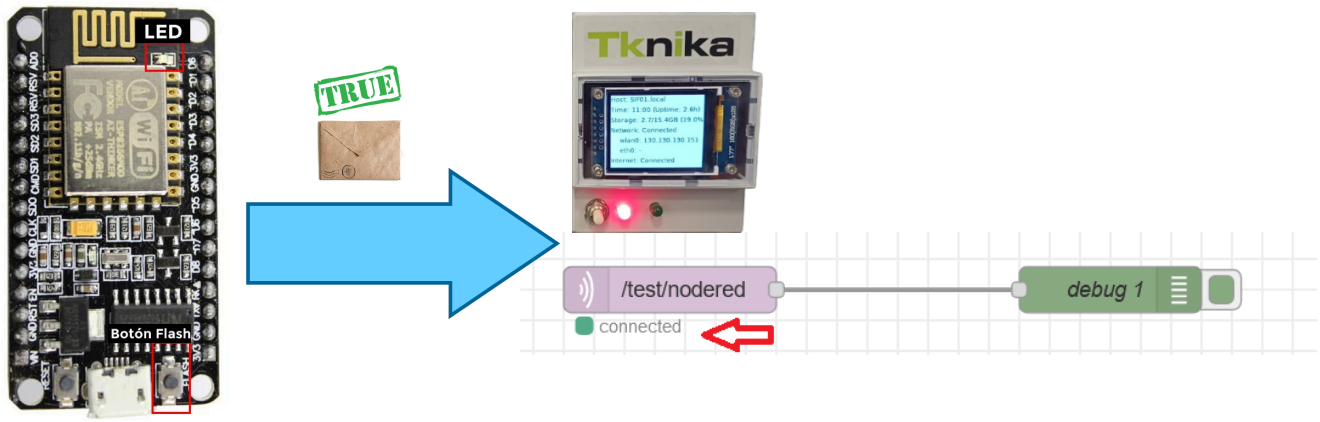


1 ROBOTIKA: FLASH botoia sakatzean TRUE mezu bat bidali (MQTT)

1 Robotikako ikasleek FLASH botoia sakatuz osatu behar duten bideojokorako erabilgarria den programa. FLASH botoia sakatzean MQTT bitartez TRUE mezua bidaliko da IoMBianera.

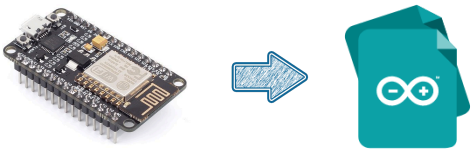
https://github.com/axpirina/NODEMCU-programak/blob/main/TRUE-MQTT.ino






```
const char* topic = "1";

if (client.connect("NodeMCUClient1", mqtt_user, mqtt_pass)) {
```

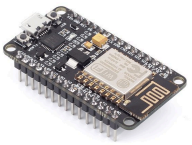
Bi lerro hauetan **1** eta **2** eta **3** NodeMCU-ak bere zenbakiekin programatu!



1 Diseinua: FLASH botoiaren egoera THINGSBOARD-era igo

Hurrengo programak NODEMCU-ko FLAS botoiaren egoera THINGSBOARD-era igotzen du.

```
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
// ===== WIFI =====
const char* ssid = "OLP";
const char* password = "oteitzaLP";
// ===== THINGBOARD =====
const char* mqtt_server = "thingsboard.cloud";
const char* access_token = "Sortutako acces oken-a idatzi";
// ===== PINAK =====
#define LED_PIN 2
#define BUTTON_PIN 0
WiFiClient espClient;
PubSubClient client(espClient);
bool lastState = false;
// ================= WIFI =================
void setup_wifi() {
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
  }
}
// ================= MQTT =================
void reconnect() {
  while (!client.connected()) {
    if (!client.connect("NodeMCU_Client", access_token, NULL)) {
      delay(5000);
    }
  }
}
// ================= TELEMETRIA =================
void sendTelemetry(bool state) {
  String payload = "{\"led_state\":";
  payload += (state ? "true" : "false");
  payload += "}";

  client.publish("v1/devices/me/telemetry", payload.c_str());
}
```



```
// ================= SETUP =================
void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT_PULLUP);
  // LED hasieran itzalita (active LOW)
  digitalWrite(LED_PIN, HIGH);
  setup_wifi();
  client.setServer(mqtt_server, 1883);
}

// ================= LOOP =================
void loop() {
  if (!client.connected()) {
    reconnect();
  }
  client.loop();

  bool pressed = (digitalRead(BUTTON_PIN) == LOW);

  // Egoera aldaketa bakarrik bidali
  if (pressed != lastState) {
    delay(30); // debounce sinplea
    pressed = (digitalRead(BUTTON_PIN) == LOW);
    if (pressed != lastState) {
      lastState = pressed;
      // LED fisikoa kontrolatu (active LOW)
      digitalWrite(LED_PIN, pressed ? LOW : HIGH);
      // Dashboard-era bidali
      sendTelemetry(pressed);
    }
  }
}
```